ARATHON RESEARCH BRIEF

Rural COTS RAN Measurement Dataset

Three high-impact research questions built around UE-side measurements of radio quality, mobility, latency, and throughput

> **Opportunity** Turn a roughly five-mile rural UE measurement campaign into explainable models, predictive maps, early-warning systems, and practical network-planning tools.

Dataset at a glance

Setting Rural commercial off-the-shelf radio access network; measurements collected from a user equipment device around an Ericsson base station.

Spatial scope Approximately five miles around the base station, according to the dataset.

Dataset Timestamped records from March 19 & 20, 2026

Measurements GPS position, latency, serving radio identifiers, radio quality indicators, and observed uplink/downlink rates.

1. Dataset description

The dataset is a geospatial time series captured at the UE while operating on a COTS RAN in a rural environment. Each row combines where and when a measurement was taken with radio-link observations and, when available, application-facing performance (throughput) tests. This combination supports research that links physical-layer conditions to user experience.

Data fields

Field group	Columns	Research value
Time and position	timestamp_local, lat, lon	Ordering, route reconstruction, speed estimation, spatial interpolation
Service experience	ping_ms, uplink, downlink	Latency and throughput outcomes visible to applications
Serving radio context	band, cellid, arfcn	Grouping by serving cell, spectrum layer, and channel
Radio quality	rsrp, sinr, rsrq	Signal strength, signal quality, and interference/loading indicators

Choosing among the three challenges

Best all-around hackathon Challenge 1 offers the clearest foundation, a strong map-based demonstration, and reusable predicted service surfaces for the other challenges.

Best research and live demonstration Challenge 2 combines explainable diagnosis, rural uplink awareness, and early warning in a route-replay experience tied directly to adaptive applications.

Best planning impact Challenge 3 translates analysis into an optimization decision, provided deployment assumptions and uncertainty are clearly stated.